IP Addressing and Subnetting

(Modified version of slides from Karim El Defrawy)

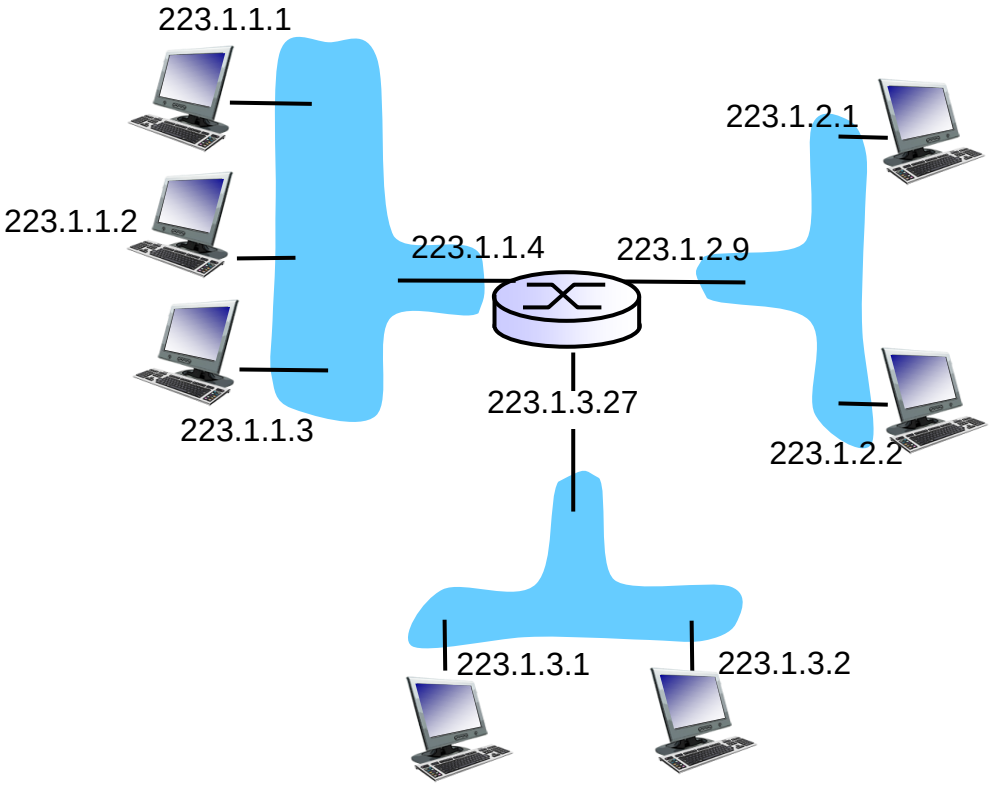
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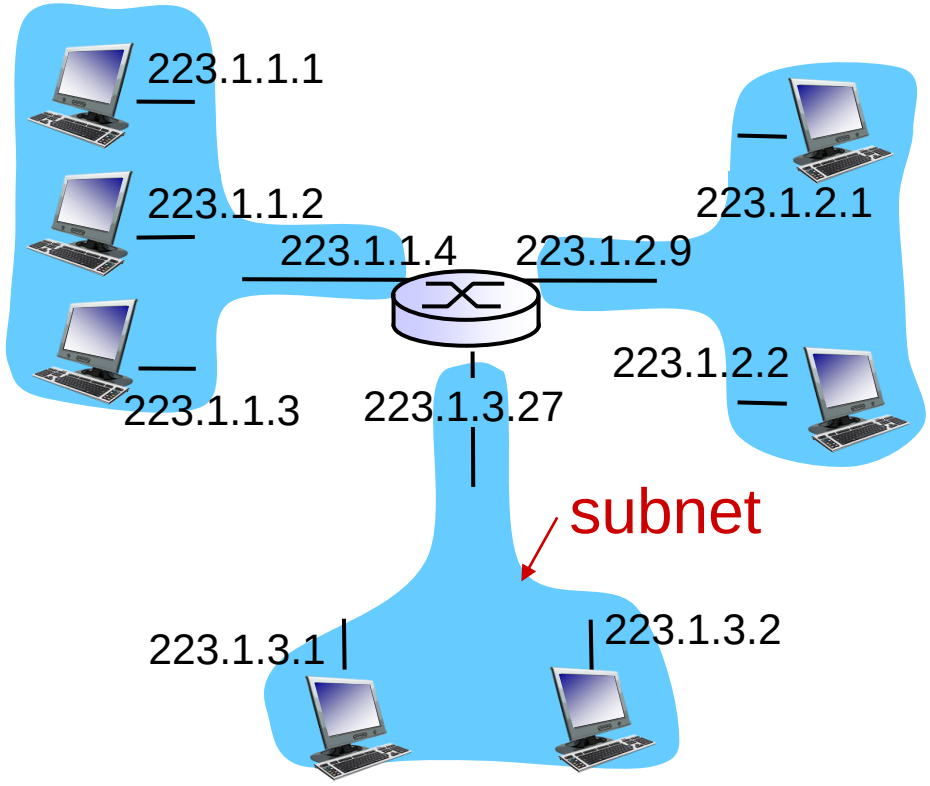
IP Addressing (IPv4)

- *IP address:* 32-bit identifier for host, router *interface*
- *interface:* connection between host/router and physical link
  - router's typically have multiple interfaces
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)
- *IP addresses associated with each interface*
- *Switches (layer 2) do not have IPs associated with interfaces*



Subnets

- IP address:
  - –subnet part - high order bits
  - –host part - low order bits
- *what's a subnet ?*
  - –device interfaces with same subnet part of IP address
  - –can physically reach each other *without intervening router*

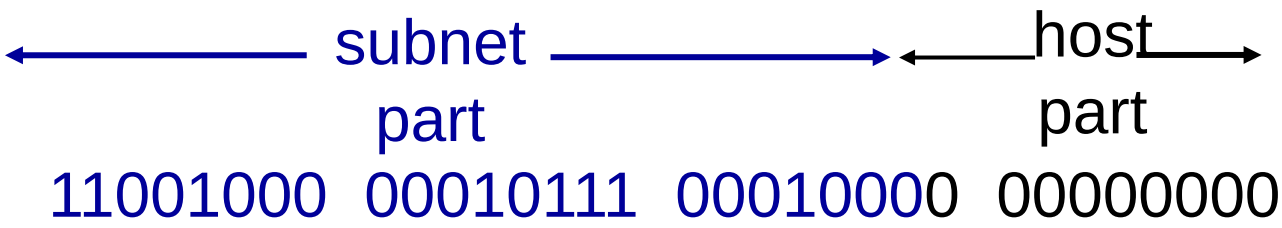


network consisting of 3 subnets

IP Addressing: CIDR

CIDR: Classless InterDomain Routing

- subnet portion of address of arbitrary length
- address format: a.b.c.d/x, where x is # bits in subnet portion of address



200.23.16.0/23

Network Address and Broadcast Address

- Network Address is the first address of the IP block
  - E.g. host IP: 10.10.1.22/24
  - Network address is: 10.10.1.0

- Broadcast Address is the last address of the IP block
  - E.g. Host IP: 10.10.1.22/24
  - Broadcast address is: 10.10.1.255

Private IP Blocks

- The Internet Assigned Numbers Authority (IANA) has assigned several address ranges to be used by private networks.

- Class A: 10.0.0.0 to 10.255.255.255
- Class B: 172.16.0.0 to 172.31.255.255
- Class C: 192.168.0.0 to 192.168.255.255

- An IP address within these ranges is therefore considered non-routable, as it is not unique can use without any coordination with IANA or an Internet registry. Addresses within this private address space are only unique within a given private network.
- All addresses outside these ranges are considered public.

More Information on Subnetting

Before subnetting:

- In any network (or subnet) one can use most of the IP addresses for host addresses.

- One loses two addresses for every network or subnet.

1. Network Address - One address is reserved to that of the network.
2. Broadcast Address – One address is reserved to address all hosts in that network or subnet.

Subnet Example

Network address **172.19.0.0 with /16 network mask**

Network	Network	Host	Host
172	19	0	0

Subnet Example

Network address **172.19.0.0 with /16 network mask**

Network	Network	Host	Host
172	19	0	0

Using Subnets: subnet mask **255.255.255.0** or /24

Network	Network	Subnet	Host
Network Mask: 255.255.0.0 or /16	11111111	11111111	00000000	00000000
Subnet Mask: 255.255.255.0 or /24	11111111	11111111	11111111	00000000

- Applying a mask which is larger than the default subnet mask, will divide your network into subnets.
- Subnet mask used here is 255.255.255.0 or /24

Subnet Example

Network address **172.19.0.0 with /16 network mask**

Using Subnets: **subnet mask** 255.255.255.0 or /24

Network	Network	Subnet	Host
172	19	0	Host
172	19	1	Host
172	19	2	Host
172	19	3	Host
172	19	etc.	Host
172	19	254	Host
172	19	255	Host

Subnets

255 Subnets

$2^8 - 1$

Cannot use last subnet as it contains broadcast address

Subnet Example

Network address **172.19.0.0 with /16 network mask**

Using Subnets: **subnet mask** 255.255.255.0 or /24

Network	Network	Subnet	Host
172	19	0	0
172	19	1	0
172	19	2	0
172	19	3	0
172	19	etc.	0
172	19	254	0
172	19	255	0

Subnets Addresses

255 Subnets

$2^8 - 1$

Cannot use last subnet as it contains broadcast address

Subnet Example

Class B address **172.19.0.0 with /16 network mask**

Using Subnets: **subnet mask** 255.255.255.0 or /24

Network	Network	Subnet	Hosts		Hosts Addresses
172	19	0	1	→	254
172	19	1	1	→	254
172	19	2	1	→	254
172	19	3	1	→	254
172	19	etc.	1	→	254
172	19	254	1	→	254

172	19	255	Host

Each subnet has 254 hosts, $2^8 - 2$

Subnet Example

Network address **172.19.0.0 with /16 network mask**

Using Subnets: **subnet mask** 255.255.255.0 or /24

Network	Network	Subnet	Host
172	19	0	255
172	19	1	255
172	19	2	255
172	19	3	255
172	19	etc.	255
172	19	254	255
172	19	255	255

Broadcast Addresses

255 Subnets

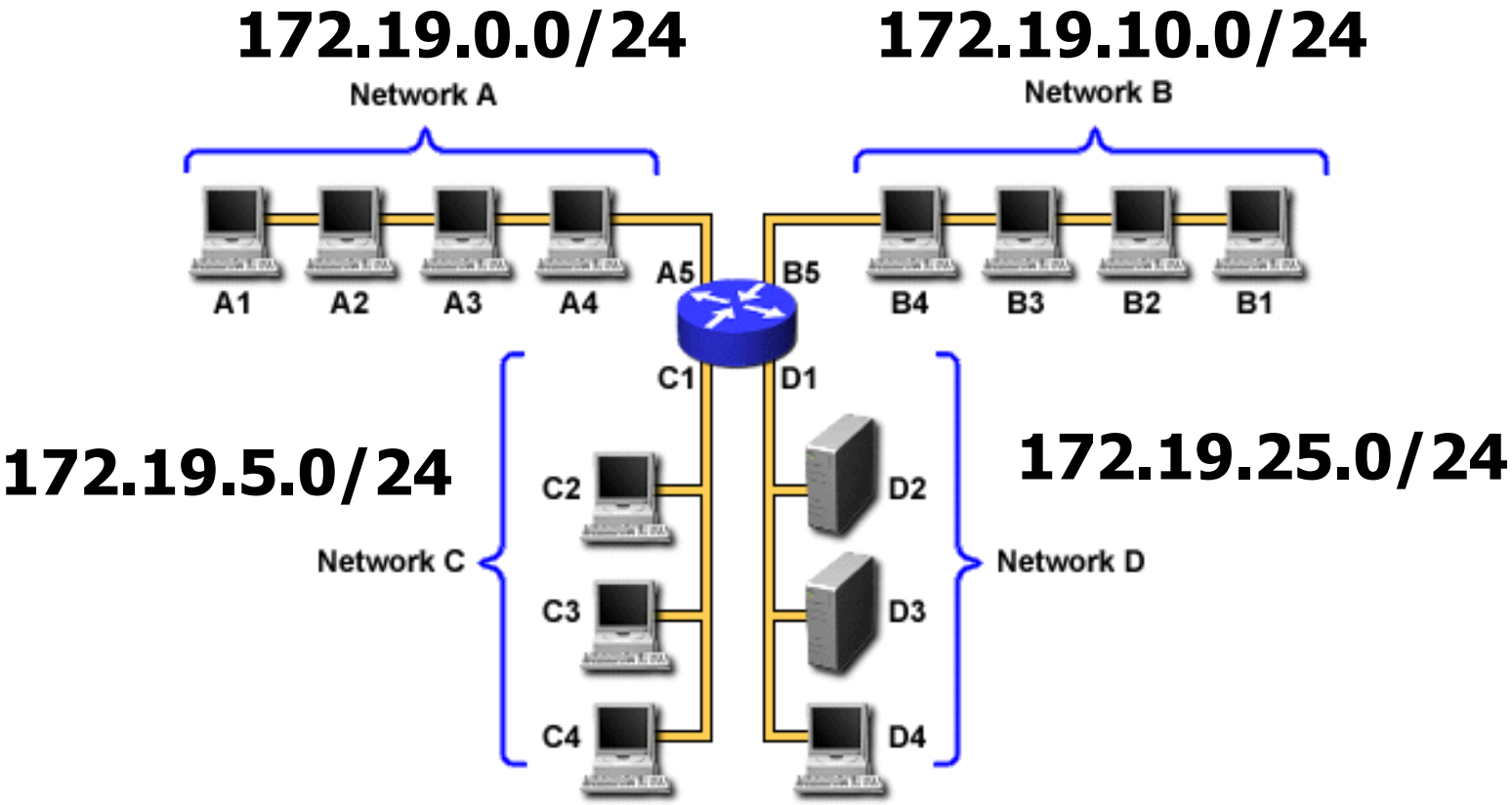
$2^8 - 1$

Cannot use last subnet as it contains broadcast address

Subnet Example

Network address **172.19.0.0 with /16 network mask**

Using Subnets: subnet mask 255.255.255.0 or /24



Important things to remember about Subnetting

- You can only subnet the host portion, you do not have control of the network portion.
- Subnetting does not give you more hosts, it only allows you to divide your larger network into smaller networks.
- When subnetting, you will actually lose host adresses:
  - For each subnet you lose the **address** of that subnet
  - For each subnet you lose the **broadcast** address of that subnet
  - **You "may" lose the first and last subnets**

- Why would you want to subnet?
  - Divide larger network into smaller networks
  - Limit layer 2 and layer 3 broadcasts to their subnet.
  - Better management of traffic.

Subnetting – Example

- **Host IP Address**: 138.101.114.250
- **Network Mask**: 255.255.0.0 (or /16)
- **Subnet Mask**: 255.255.255.192 (or /26)

Given the following Host IP Address, Network Mask and Subnet mask find the following information:

- Major Network Information
  - Major Network Address
  - Major Network Broadcast Address
  - Range of Hosts if not subnetted
- Subnet Information
  - Subnet Address
  - Range of Host Addresses (first host and last host)
  - Broadcast Address
- Other Subnet Information
  - Total number of subnets
  - Number of hosts per subnet

Major Network Information

- **Host IP Address**: 138.101.114.250
- **Network Mask**: 255.255.0.0
- **Subnet Mask**: 255.255.255.192

- **Major Network Address**: 138.101.0.0
- **Major Network Broadcast Address**: 138.101.255.255
- **Range of Hosts if not Subnetted**: 138.101.0.1 to 138.101.255.254

Step 1: Convert to Binary

128	64	32	16	8	4	2	1

	138.	**101.**	**114.**	**250**
IP Address	10001010	01100101	01110010	11111010
Mask	11111111	11111111	11111111	11000000
	255.	**255.**	**255.**	**192**

Step 1:

Translate Host IP Address and Subnet Mask into binary notation

Step 2: Find the Subnet Address

	138.	101.	114.	250
IP Address	10001010	01100101	01110010	11111010
Mask	11111111	11111111	11111111	11000000
Network	10001010	01100101	01110010	11000000
	138	**101**	**114**	**192**

Step 2:

Determine the Network (or Subnet) where this Host address lives:

1. Draw a line under the mask
2. Perform a bit-wise AND operation on the IP Address and the Subnet Mask

   Note: 1 AND 1 results in a 1, 0 AND anything results in a 0
3. Express the result in Dotted Decimal Notation
4. The result is the **Subnet Address** of this Subnet or “Wire” which is 138.101.114.192

Step 2: Find the Subnet Address

	138.	101.	114.	250
IP Address	10001010	01100101	01110010	11111010
Mask	11111111	11111111	11111111	11000000
Network	10001010	01100101	01110010	11000000
	138	**101**	**114**	**192**

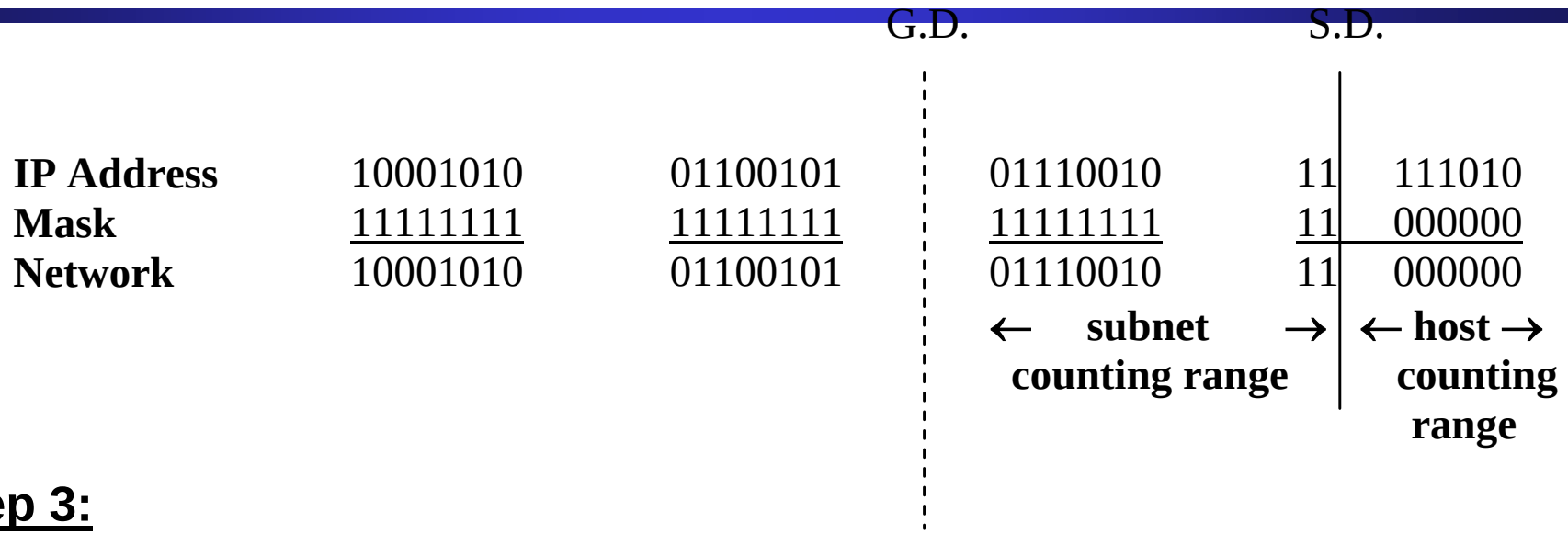
Step 2:

Determine the Network (or Subnet) where this Host address lives:

Quick method:

1. Find the last (right-most) 1 bit in the subnet mask.
2. Copy all of the bits in the IP address to the Network Address
3. Add 0's for the rest of the bits in the Network Address

Step 3: Subnet Range / Host Range

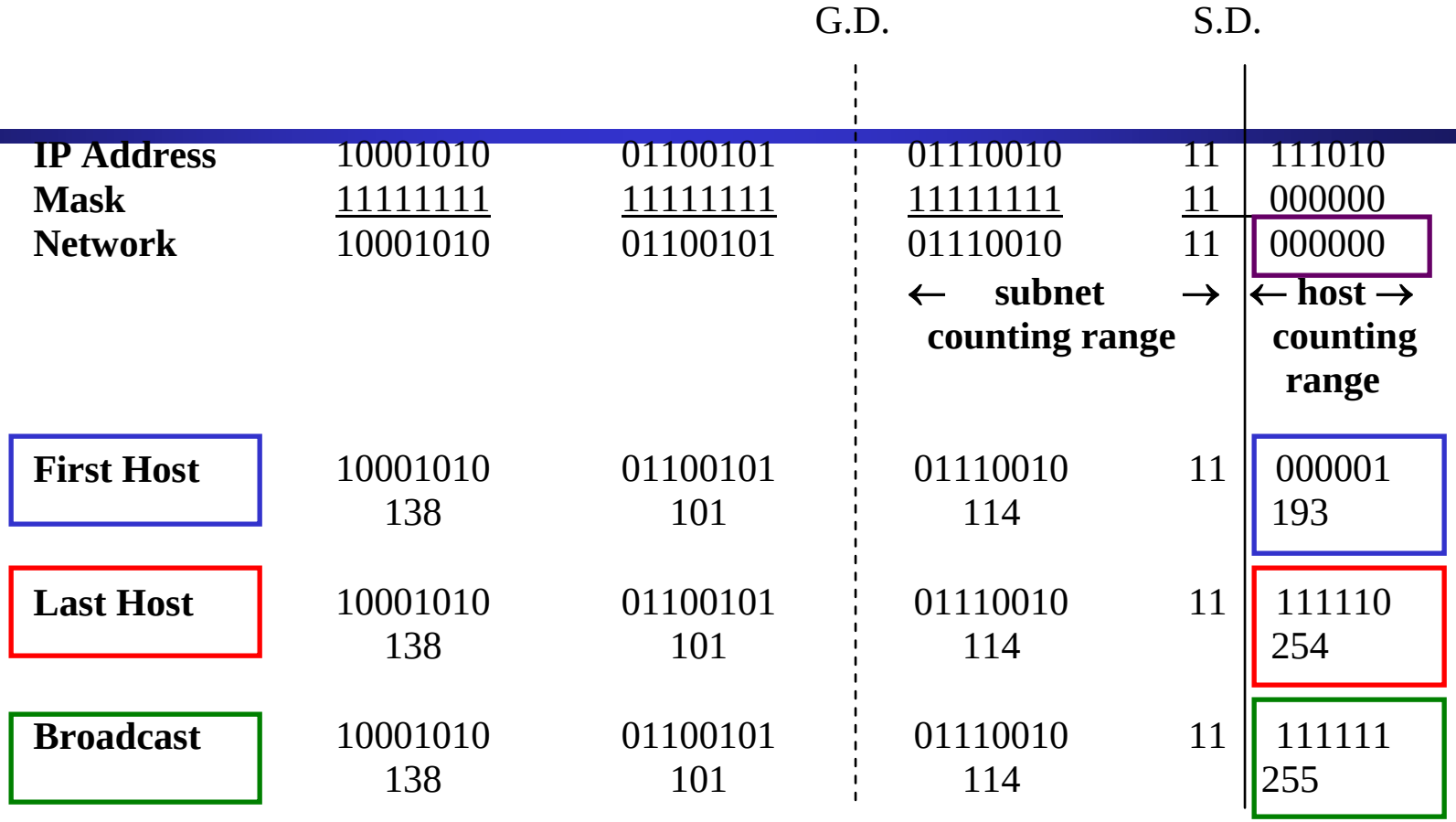


Step 3:

Determine which bits in the address contain Network (subnet) information and which contain Host information:

- Use the **Network Mask**: 255.255.0.0 and divide (**Great Divide**) the from the rest of the address.
- Use **Subnet Mask**: 255.255.255.192 and divide (**Small Divide**) the subnet from the hosts between the last “1” and the first “0” in the subnet mask.

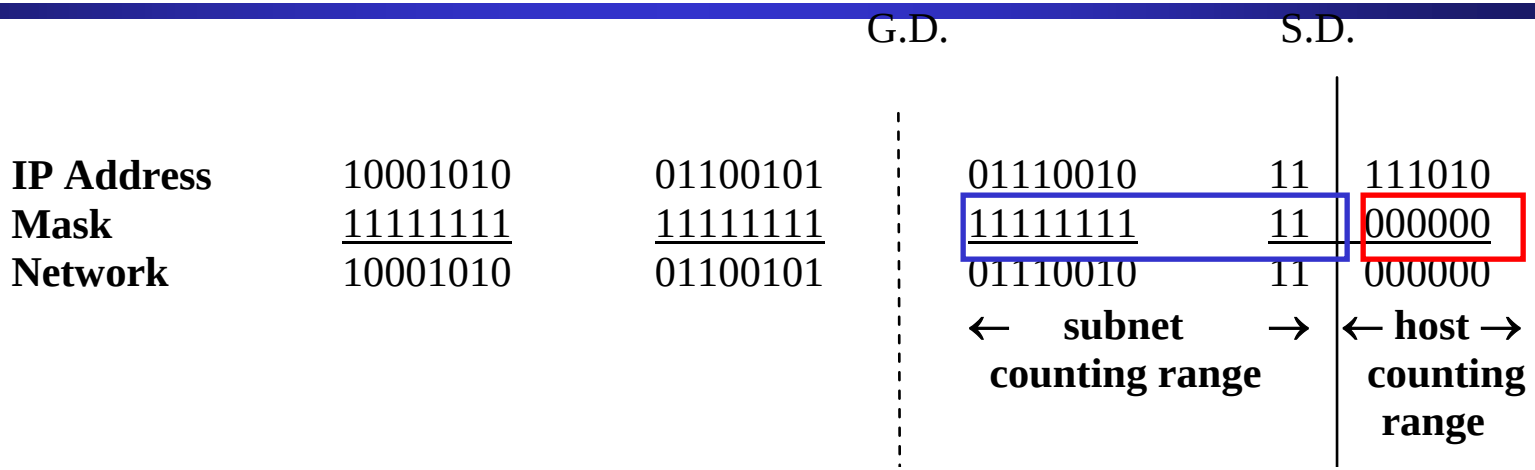
Step 4: First Host / Last Host



Host Portion

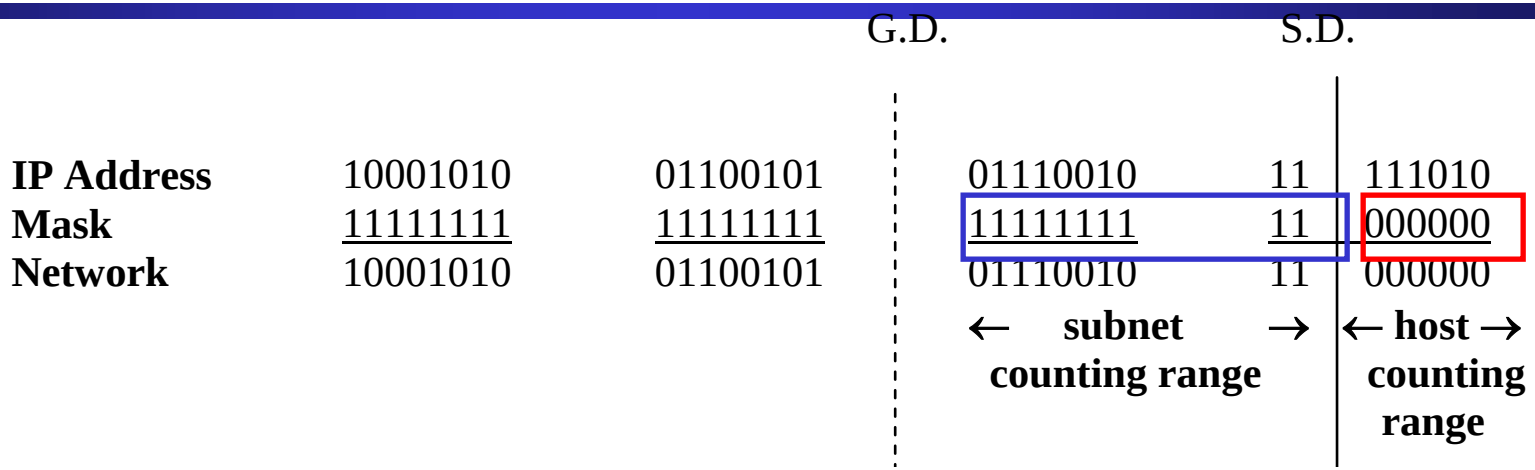
- **Subnet Address:** all 0's
- **First Host**: all 0's and a 1 in rightmost bit
- **Last Host**: all 1's and a 0 in rightmost bit
- **Broadcast**: all 1's

Step 5: Total Number of Subnets



- Total number of subnets
  - Number of subnet bits 10
  - $2^{10} = 1{,}024$
  - 1,024 total subnets
    - Subtract one “**if**” all-zeros subnet cannot be used
    - Subtract one “**if**” all-ones subnet cannot be used

Step 6: Total Number of Hosts per Subnet



- Total number of hosts per subnet
  - Number of host bits 6
  - $2^6 = 64$
  - 64 host per subnets
    - Subtract one for the subnet address
    - Subtract one for the broadcast address
  - 62 hosts per subnet